

Department Overview • Department Name: Department of Textile Engineering & Design • Faculty: Faculty of Engineering & Architecture • Accreditation: Fully accredited by the Pakistan Engineering Council (PEC). • Core Focus: Encompasses the full manufacturing lifecycle of textiles-transforming raw fibers into yarn, spinning yarn into fabric, and executing garment/apparel and home textile designs. It includes processing of wool, cotton, and manmade fibers, technical textiles, color science, and environmental sustainability in manufacturing. • History & Impact: Launched in 2004 as a pioneer engineering program at the Takatu Campus with an initial intake of 20 students. Strategically positioned because Balochistan is rich in raw wool production. The department introduced its postgraduate (MS) program in 2015. • Career Pathways: Graduates work as technical experts, supervisors, and managers across textile processing plants, quality control units, research labs, product development divisions, and corporate management. The field also branches into specialized biomedical applications (e.g., medical filters and artificial arteries).

Vision and Mission Vision: To be recognized internationally for its quality research, educational activities, and its relationship with the environment. Its graduates will be acknowledged for the strength of their technical and practical knowledge, ability to work as a team, and their vocation of service to society. Mission: To provide professional training, promoting social responsibility, values, and a progressive research culture among the students of Textile Engineering, making them capable of providing solutions to complex engineering problems, preserving the environment, and contributing to the sustainable development of Pakistan and the society.

1. BS Textile Engineering (Engineering Track)

Admission Requirements • Academic Background: F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from a recognized board (or equivalent qualification) with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in a relevant discipline with a minimum of 60% marks. • Testing: Success in the NTS Test is required.

Degree Requirements • Total Credit Hours: 136 • Total Courses: 50 • Minimum Academic Standing: CGPA $\geq$ 2.0

Program Educational Objectives (PEOs)

1. Implement real-world applications making use of fundamental engineering knowledge, designs, and managerial skills gained during the textile engineering program.
2. Generate innovative and sustainable solutions to complex engineering problems contributing to the textile industry.
3. Promote themselves through communication and interpersonal skills, career management strategies, and the ability to deal with technological, socio-economical, environmental, and ethical challenges.

Curriculum Plan (Program Schema by Semester) Note: Credit hour listings follow the format (Theory - Lab).

1st Semester • MATHP-212 / CHE-102 & L | Linear Algebra & Geometry / Applied Chemistry | Prerequisite: None / None | Credit Hours: 3-0 / 3-1 • PHY-205 & L / TXTE-202 | Applied Physics / Textile Raw Materials-I | Prerequisite: None / None | Credit Hours: 3-1 / 3-0 • HUM-163 / TXTE-103 | Functional English (English-I) / Intro to Textile Eng. | Prerequisite: None / None | Credit Hours: 2-0 / 2-0

2nd Semester • MATHP-167 / ENGG-201L | Calculus-I / Engineering Drawing & Graphics |

Prerequisite: None / None | Credit Hours: 2-0 / 0-1 • ENGG-218 & L / TXTE-313 | Engineering Mechanics / Textile Raw Materials-II | Prerequisite: None / None | Credit Hours: 2-1 / 3-0 • HUM-101 / CS-105TXTE & L | Islamic Studies / Introduction to Computing | Prerequisite: None / None | Credit Hours: 2-0 / 1-1 • HUM-268 / MATHA-546 & L | Communication Skills (Eng-II) / Thermodynamics | Prerequisite: None / None | Credit Hours: 2-0 / 2-1

3rd Semester • MATHP-267 / TXTE-401 & L | Calculus-II / Fibre Science | Prerequisite: None / None | Credit Hours: 2-0 / 3-1 • ENGG-219 / HUM-102 | Machine Design / Pakistan Studies | Prerequisite: None / None | Credit Hours: 2-0 / 2-0 • TXTE-302 & L / TXTE-231 & L | Spinning Preparatory Process / Fabric Manufacturing-I | Prerequisite: None / None | Credit Hours: 2-1 / 2-1

4th Semester • STAT-103 / EE-209 & L | Statistics and Probability / Electrical & Electronic Eng. | Prerequisite: None / None | Credit Hours: 3-0 / 2-1 • HUM-207 | Professional Ethics | Prerequisite: None | Credit Hours: 2-0 • TXTE-321 & L / TXTE-331 & L | Yarn Production Process / Fabric Manufacturing-II | Prerequisite: None / None | Credit Hours: 3-1 / 3-1

5th Semester • MGMT-304 / HUM-265 | Organizational Behavior / Tech Writing & Pres. Skills | Prerequisite: None / Eng-I | Credit Hours: 2-0 / 3-0 • TXTE-422 & L / TXTE-431 & L | Spinning Planning & Production / Weave Designs | Prerequisite: None / None | Credit Hours: 2-1 / 3-1 • TXTE-304 & L / TXTE-124 & L | Pre-treatment in Textile Wet Processing / Garment Fund. | Prerequisite: None / None | Credit Hours: 2-1 / 2-1

6th Semester • MGMT-314 / EE-343 & L | Operations & Project Management / Process Systems | Prerequisite: None / None | Credit Hours: 2-0 / 2-1 • TXTE-456 & L | Textile Dyes & Dyeing | Prerequisite: Applied Chemistry | Credit Hours: 2-1 • TXTE-455 & L / TXTE-453 | Anthropometry & Clothing Construction / Tech Textiles | Prerequisite: None / None | Credit Hours: 2-1 / 2-0 • TXTE-311 / TXTE-402 | Textile Utilities & Services / Color Science | Prerequisite: None / None | Credit Hours: 2-0 / 2-0 • Mandatory | Industry Internship (4-6 weeks) | Non-Credit

7th Semester • ECON-105 / TXTE-452 | Engineering Economics / Merchandizing | Prerequisite: None / None | Credit Hours: 2-0 / 2-0 • TXTE-441L / CS-203L | Design Project-I / Computer Application in Textiles | Prerequisite: None / Computing | Credit Hours: 0-3 / 0-2 • TXTE-351 & L / TXTE-352 & L | Mechanism of Garment Machines / Sewn Production Eng. | Prerequisite: None / None | Credit Hours: 2-1 / 2-1 • TXTE-432 | Denim Technology | Prerequisite: None | Credit Hours: 2-0

8th Semester • MGMT-201 / TXTE-444 & L | Entrepreneurship / Textile Testing & Quality Control | Prerequisite: None / None | Credit Hours: 2-0 / 3-1 • ENGG-323 & L | Environmental Issues & Control | Prerequisite: None | Credit Hours: 2-1 • TXTE-457 & L / TXTE-442L | Textile Printing & Finishing / Design Project-II | Prerequisite: None / None | Credit Hours: 3-1 / 0-3

2. BS Fashion & Textile Design (Non-Engineering Track)

Admission Requirements • Academic Background: Intermediate certification (F.Sc./ICS/F.A.) from any recognized board or an equivalent qualification with at least 45% marks. • Testing: Must pass both the central NTS Test and a localized Free Hand Drawing Test.

Degree Requirements • Total Credit Hours: 136 • Total Courses: 64 • Minimum Academic Standing: CGPA $\geq$ 2.0